Homework 5

Stephen Gibbs

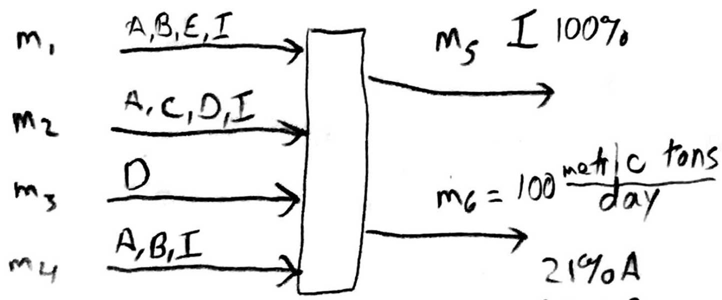
1) A)

wt%

| | $m_1$ Limestone | $m_2$ Clay | $m_3$ Millscale | $m_4$ Oyster shells |
|---|---|---|---|---|
| $A = SiO_2$ | 1 | 68 | 0 | 2.1 |
| $B = CaO$ | 54 | 0 | 0 | 55 |
| $C = Al_2O_3$ | 0 | 20 | 0 | 0 |
| $D = FeO_3$ | 0 | 4 | 100 | 0 |
| $E = MgO$ | 4 | 0 | 0 | 0 |
| $I$ = Ignitables | 41 | 8 | 0 | 42.9 |

1) 17
2) 20
3) 10
4) 20
5) 20

87/100




unks: $m_1, m_2, m_3, m_4, m_5, W_{6C}, W_{6E}$

$21 \frac{\text{m tons}}{\text{day}}$

A: $m_1 \cdot 0.01 + m_2 \cdot 0.68 + m_4 \cdot 0.021 = 100(0.21)$

B: $m_1 \cdot 0.54 + m_4 \cdot 0.55 = 100(0.65) = 65 \frac{\text{m tons}}{\text{day}}$

C: $m_2 \cdot 0.20 = W_{6C}(100)$

D: $m_2 \cdot 0.04 + m_3 = W_{6D} \cdot 100$

E: $m_1 \cdot 0.04 = W_{6E} \cdot 100$

I: $m_1 \cdot 0.41 + m_2 \cdot 0.08 + m_4 \cdot 0.429 = m_5$

TOT: $m_1 + m_2 + m_3 + m_4 = m_5 + 100 \frac{\text{mtons}}{\text{day}}$

eqn | unk | solved on excel

$$
\begin{array}{c}
\\ A \\ B \\ C \\ D \\ E \\ I \\ TOT
\end{array}
\begin{array}{c}
\begin{array}{ccccccc} m_1 & m_2 & m_3 & m_4 & m_5 & W_{6C} & W_{6E} \end{array} \\
\begin{pmatrix}
0.01 & 0.68 & 0 & 0.021 & 0 & 0 & 0 \\
0.54 & 0 & 0 & 0.55 & 0 & 0 & 0 \\
0 & 0.20 & 0 & 0 & 0 & -100 & 0 \\
0 & 0.04 & 1 & 0 & 0 & 0 & 0 \\
0.04 & 0 & 0 & 0 & 0 & 0 & -100 \\
0.41 & 0.08 & 0 & 0.429 & -1 & 0 & 0 \\
1 & 1 & 1 & 1 & -1 & 0 & 0
\end{pmatrix}
\end{array}
\begin{pmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \\ W_{6C} \\ W_{6E} \end{pmatrix}
=
\begin{pmatrix} 21 \\ 65 \\ 0 \\ 5 \\ 0 \\ 0 \\ 100 \end{pmatrix}
$$

$m_1 = 102.9$
$m_2 = 28.8$
$m_3 = 2.96$
$m_4 = 17.2$
$m_5 = 51.86$
$W_{6C} = 0.057$
$W_{6E} = 0.033$

18/20

B)

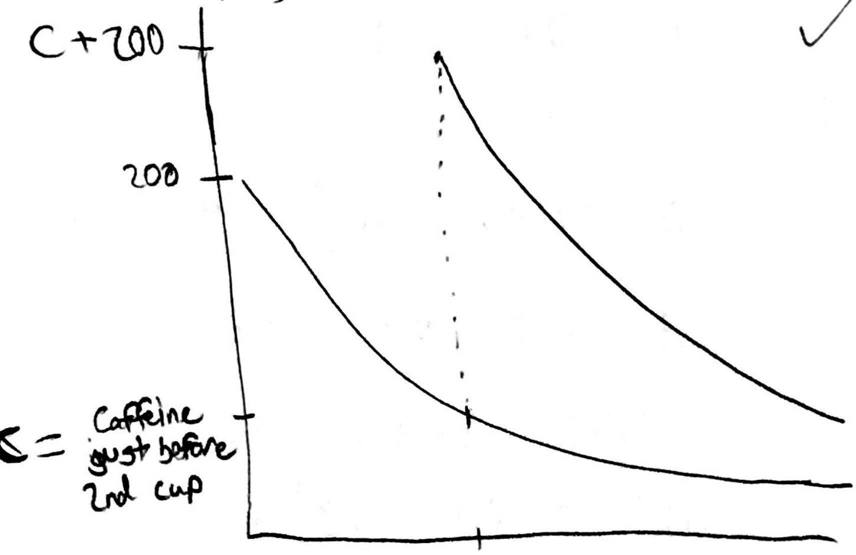
| eqn \ unks | $m_1$ | $m_2$ | $m_3$ | $m_4$ | $m_5$ | $W_{GC}$ | $W_{GE}$ |
|---|---|---|---|---|---|---|---|
| 1 | ✓ | ✓ | | ✓ | | 0 | 0 |
| 2 | ✓ | | | ✓ | | 0 | 0 |
| 3 | | ✓ | | | | ✓ | |
| 4 | | ✓ | ✓ | | | 0 | |
| 5 | ✓ | | | | | | ✓ |
| 6 | ✓ | ✓ | | ✓ | ✓ | 0 | 0 |
| 7 | ✓ | ✓ | ✓ | ✓ | ✓ | 0 | 0 |

Successive substitution would not work because you always have more unks than equations with matching unks, so there is no way to simplify it.

2) DOF: $(2+1)-(1+1+1)=0$

$$\frac{dm_{,sys}}{dt} = -0.116 m_{c,sys} = -\dot{m}_{c,out} \Rightarrow \int_{200}^{100} \frac{1}{m_{c,sys}}\, dm_{c,sys} = \int_0^{t_f} -0.116\, dt$$

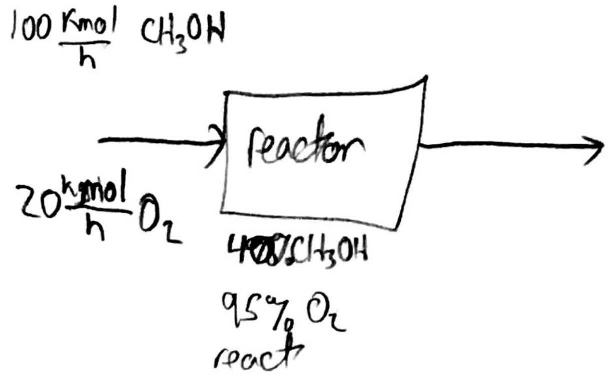
$$\ln\left(\frac{100}{200}\right) = -0.116(t_f) \Rightarrow t_f = 5.98 \text{ hrs}$$




Caffeine content would be significantly higher with the second cup.

$\frac{20}{20}$

3)




$CH_3OH = M$
$O_2 = O$
$HCHO = F$
$HCOOH = A$
$CO$
$CO_2$
$H_2O = W$

DOF: $(9+4)-(2+0+2+7)$

$13 - 11 = 2$ ✓

$CH_3OH + \frac{1}{2}O_2 \rightarrow HCHO + H_2O$

$CH_3OH + O_2 \rightarrow HCOOH + H_2O$

$CH_3OH + O_2 \rightarrow CO + 2H_2O$

$CH_3OH + \frac{3}{2}O_2 \rightarrow CO_2 + 2H_2O$

$\dot{n}_{M,out} = 0.6\,\overset{100}{\dot{n}_{M,in}} = 60\,\frac{kmol}{hr}$

$\dot{n}_{O,out} = 0.05\,\overset{20}{\dot{n}_{O,in}} = 1\,\frac{kmol}{hr}$

21.5 ✓

$M: 0 = 100\frac{kmol}{hr} - 60\frac{kmol}{hr} + 0 - (\xi_1 + \xi_2 + \xi_3 + \xi_4) \Rightarrow \xi_1 + \xi_2 + \xi_3 + \xi_4 = 40\frac{kmol}{hr}$

$O: 0 = 20\frac{kmol}{hr} - 1\frac{kmol}{hr} + 0 - (\frac{1}{2}\xi_1 + \xi_2 + \xi_3 + \frac{3}{2}\xi_4) \Rightarrow \frac{1}{2}\xi_1 + \xi_2 + \xi_3 + \frac{3}{2}\xi_4 = 19\frac{kmol}{hr}$

$F: \dot{n}_{F,out} = \xi_1$

$A: \dot{n}_{A,out} = \xi_2$

$CO: \dot{n}_{CO,out} = \xi_3$

$CO_2: \dot{n}_{CO_2,out} = \xi_4$

$W: \dot{n}_{W,out} = \xi_1 + \xi_2 + 2\xi_3 + 2\xi_4$

$\frac{10}{20}$

1)


DOF: $2+2-(2+0+1+1)=0$

$acc = IN - OUT + \overset{0}{\cancel{GEN}} - cons$

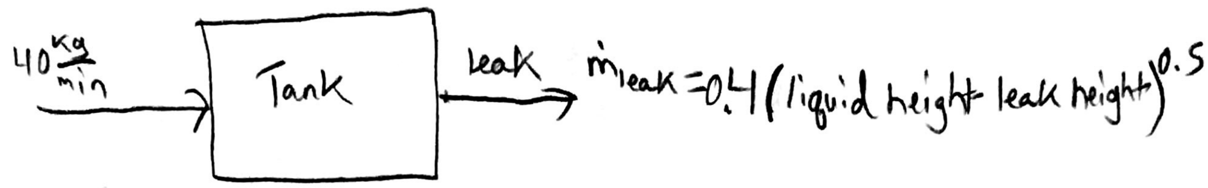
$acc = 62000\frac{T}{hr}(8hr) - 57000\frac{T}{hr}(8hr) - \int_0^8 2700(1-e^{-0.3t})\,dt$

$acc = 40{,}000 - 13416 = 26584$ T accumulated over 8 hrs

The product does not meet the required 30,000 over 8 hrs specifications.

$\frac{20}{20}$

1)




Vol at leak $= \pi(0.5m)^2 \cdot 1m = 0.785\ m^3$

$0.785m^3 \times 1000\frac{kg}{m^3} = 785\ kg$ solution, $78.5\ kg\ X$

$\overset{78.5}{\hat{m}_{x_f}} - \overset{0}{\hat{m}_{x_i}} = \int_0^{t_{leak}} \overset{4}{\dot{m}_{x_{in}}}\,dt \Rightarrow 78.5 = 4(t_{leak} - 0) \Rightarrow t_{leak} = 19.6\ min$ for solution to reach the leak

$$\frac{dm_x}{dt} = 78.5\frac{dh}{dt} = \dot{m}_{x_{in}} - \dot{m}_{x_{out}} = 4 - 0.4\sqrt{h-1}$$ ✓

$$\int_1^{h_f} \frac{78.5}{4 - 0.4\sqrt{h-1}}\,dh = \int_{19.6}^{40} dt = 40 - 19.6 = 20.4\ min$$ ✓

optimized on excel $h_f = 1.972$

$(1.972 - 1)(\pi 0.5^2)(1000)(0.1)$

$0.972m \times 0.785m^2 \times 1000\frac{kg}{m^3} \times 0.1\frac{X}{sol} = 76.3\ kg\ X$ in tank above leak

$40\frac{kg}{min}(40\ min - 19.6) \times 0.1 = 81.6\ kg$ pumped in from 19.6 min to 40 min

$81.6 - 76.3 = 5.3\ kg$ of X have leaked out ✓

20/20